

Integrated Circuit ECE326

Lec03

Synthesis Digital Circuit

Dr. Ahmed Reda Mohamed^{1,2}

¹Ph.D, the MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (*BiCASL*), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.

²Assistant professor, Department of Electronics and Communications, Zagazig University, Zagazig , Egypt.



- ☐ **Classification of transistors**
- ☐ **Enhancement MOSFET**
- ☐ **Ch/chs of E-MOSFET as a switch**
- ☐ **Synthesize of CMOS digital circuit**



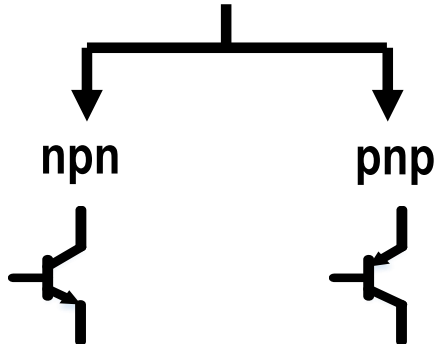
- ☐ **Classification of transistors**
- ☐ **Enhancement MOSFET**
- ☐ **Ch/chs of E-MOSFET as a switch**
- ☐ **Synthesize of CMOS digital circuit**

Classification of transistors



Transistor

Bipolar Junction Transistor
(BJT) [**Bipolar device**]



Junction FET
(JFET)

Enhancement MOSFET
(E-MOSFET)

n-channel

p-channel

Normally **“off”**



Field Effect Transistor
(FET) [**Unipolar device**]

Metal Oxide semi-conductor FET
(MOSFET)

Depletion MOSFET
(D-MOSFET)

n-channel

p-channel

Normally **“On”**

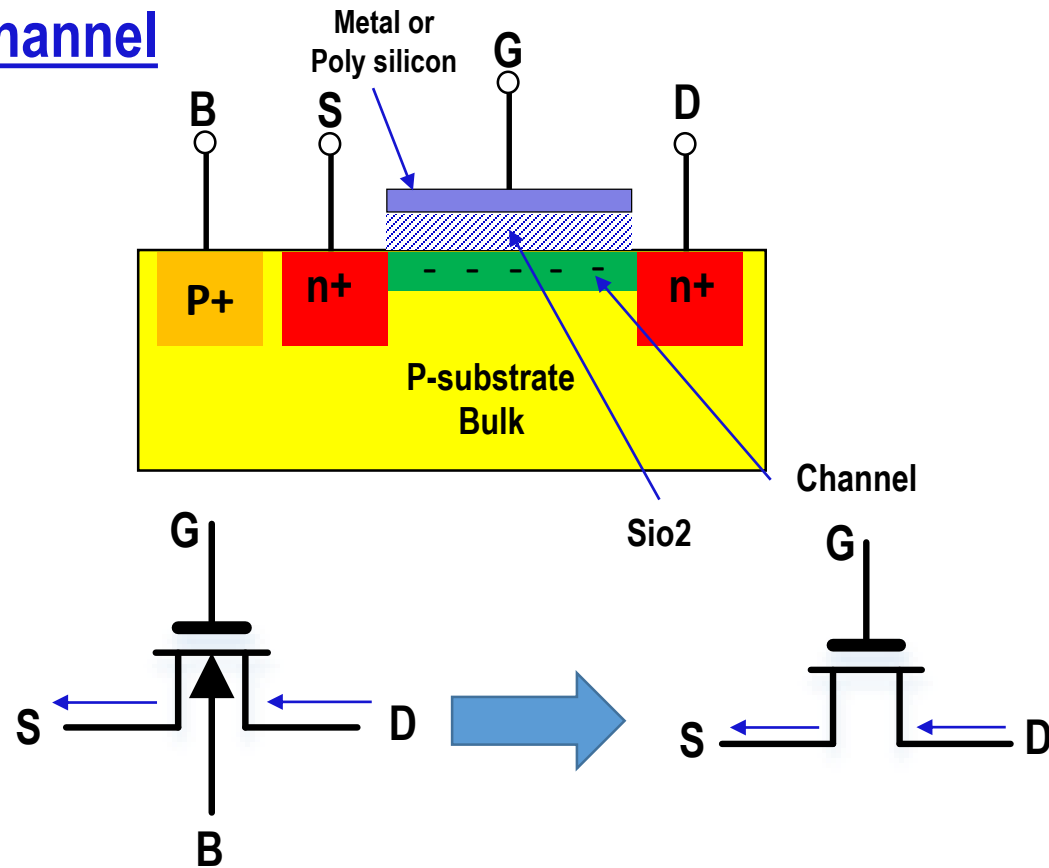




- ❑ **Classification of transistors**
- ❑ **Enhancement MOSFET**
- ❑ **Ch/chs of E-MOSFET as a switch**
- ❑ **Synthesize of CMOS digital circuit**

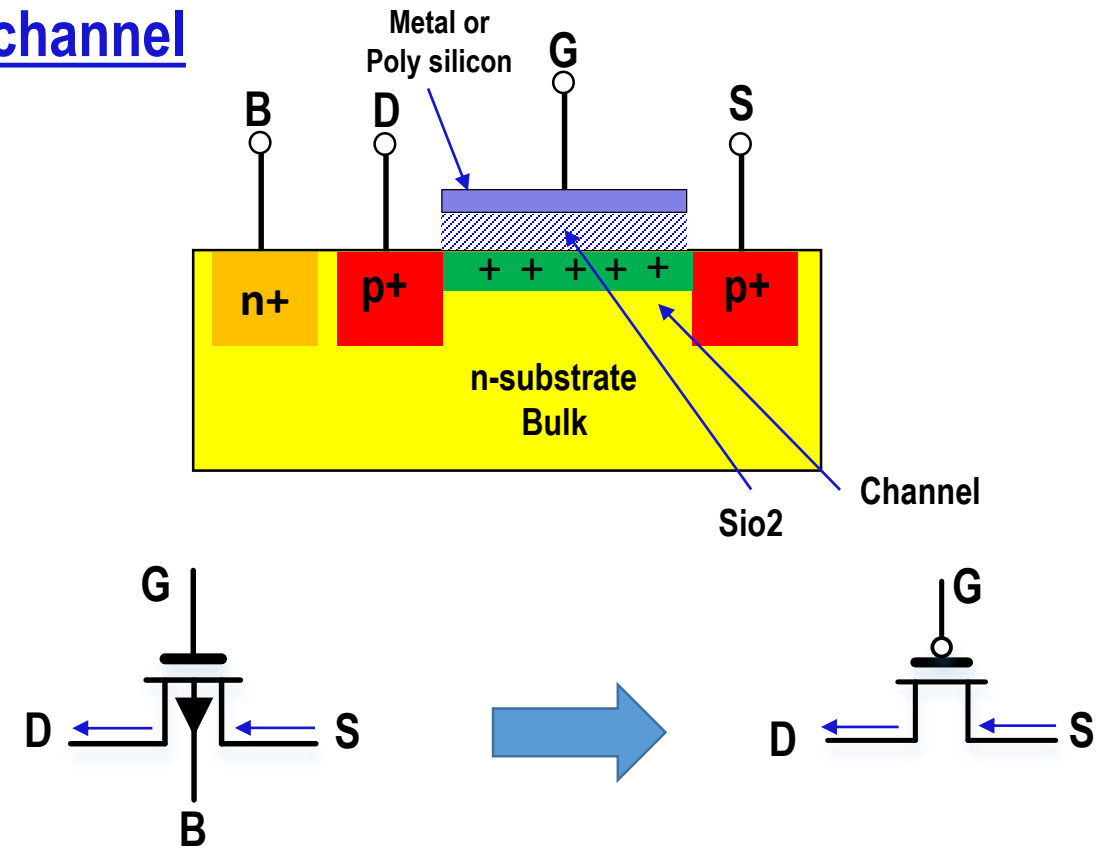
A- Construction

n-channel



- ☐ $V_D > V_S$
- ☐ Current direction from Drain to Source (I_{DS})
- ☐ $I_G = 0$

p-channel

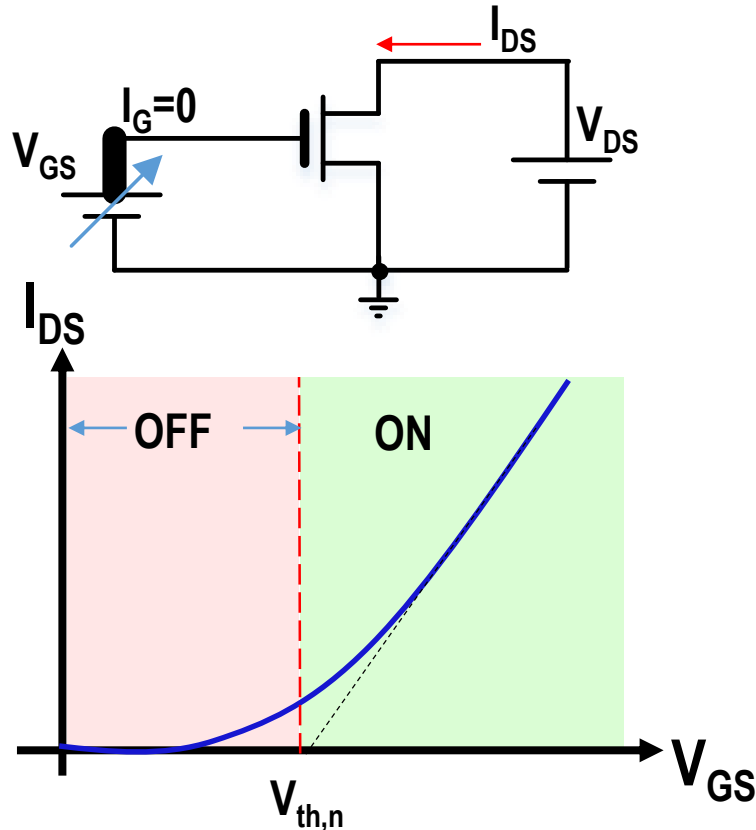


- ☐ $V_S > V_D$
- ☐ Current direction from Source to Drain ($I_{SD} = -I_{DS}$)
- ☐ $I_G = 0$

B- Input transfer curve

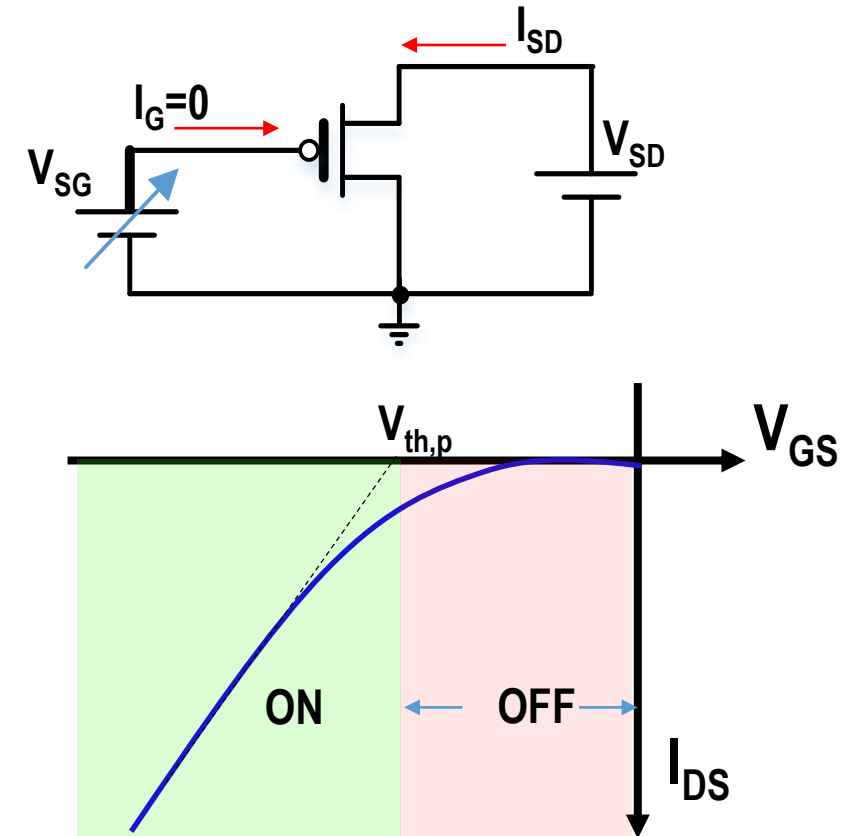
Relation between I_{DS} Vs V_{GS}

n-channel



- ☐ $V_{th,n}$: Threshold voltage of n-channel
- ☐ $V_{GS} \geq V_{th,n}$ to construct the n-channel

p-channel



- ☐ $V_{th,p}$: Threshold voltage of p-channel
- ☐ $V_{GS} \leq V_{th,p}$ to construct the p-channel

Ex: $V_{th,n} = 0.5 \text{ V}$, $V_{th,p} = - 0.5 \text{ V}$

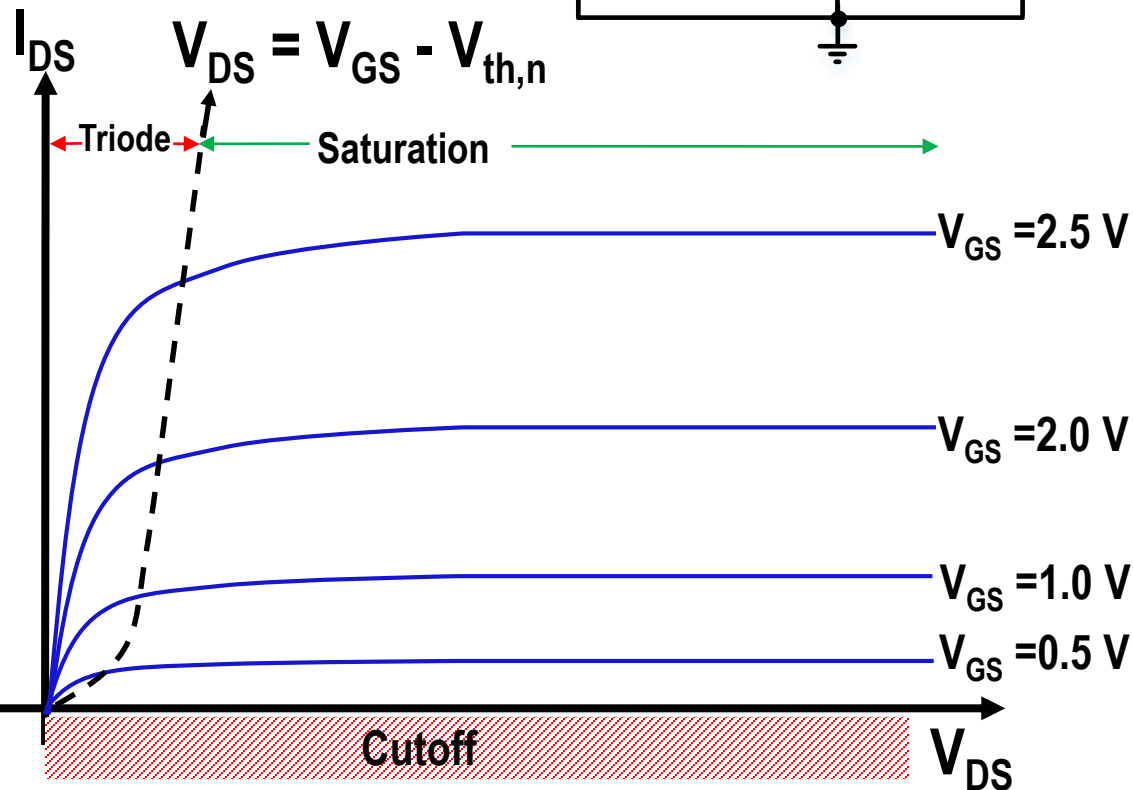
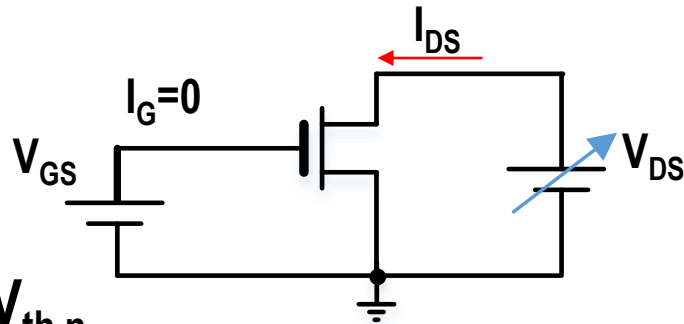
Enhancement MOSFET (3/5)



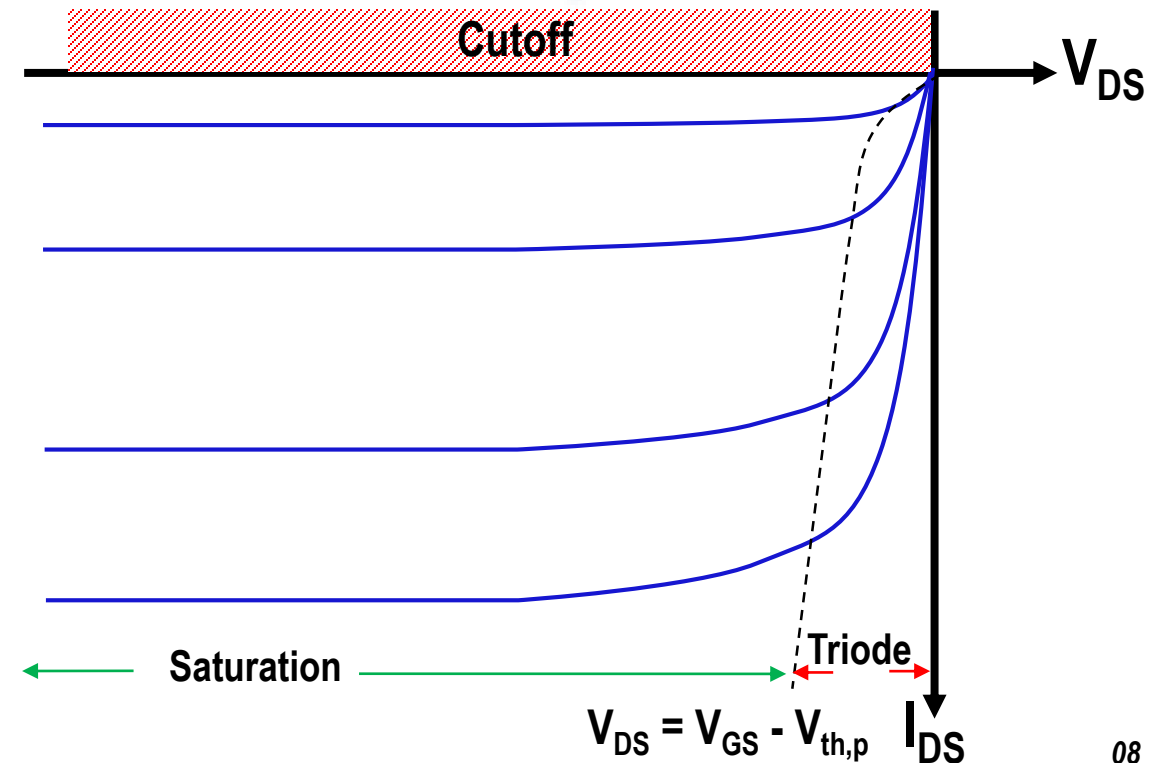
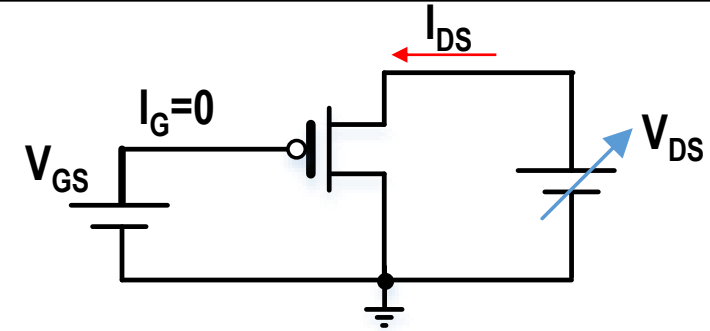
C- Output Characteristic curve

Relation between I_{DS} Vs V_{DS}

n-channel



p-channel



D-Related Equations

n-channel

❑ Cutoff

$$\square V_{GS} < V_{th,n} \quad \longrightarrow \quad I_{DS}=0$$

❑ Triode

$$\square V_{GS} \geq V_{th,n} \quad \& \quad V_{DS} < V_{GS} - V_{th,n}$$

$$\square I_{DS} = B_n \left((V_{GS} - V_{th,n}) V_{DS} - \frac{V_{DS}^2}{2} \right)$$

❑ Saturation

$$\square V_{GS} \geq V_{th,n} \quad \& \quad V_{DS} \geq V_{GS} - V_{th,n}$$

$$\square I_{DS} = \frac{B_n}{2} (V_{GS} - V_{th,n})^2$$

p-channel

❑ Cutoff

$$\square V_{GS} > V_{th,p} \quad \longrightarrow \quad I_{DS}=0$$

❑ Triode

$$\square V_{GS} \leq V_{th,p} \quad \& \quad V_{DS} > V_{GS} - V_{th,p}$$

$$\square I_{DS} = -B_p \left((V_{GS} - V_{th,p}) V_{DS} - \frac{V_{DS}^2}{2} \right)$$

❑ Saturation

$$\square V_{GS} \leq V_{th,p} \quad \& \quad V_{DS} \leq V_{GS} - V_{th,p}$$

$$\square I_{DS} = -\frac{B_p}{2} (V_{GS} - V_{th,p})^2$$

$$B_n = \mu_n C_{ox} \frac{W}{L} \quad \& \quad B_p = \mu_p C_{ox} \frac{W}{L}$$

$$C_{ox} = \frac{\epsilon_0 \epsilon_r}{t_{ox}}$$

B_n : Gain process factor n-MOSFET

μ_n : Mobility of electron

C_{ox} : Gate Oxide Capacitor/unit area

W : width of transistor

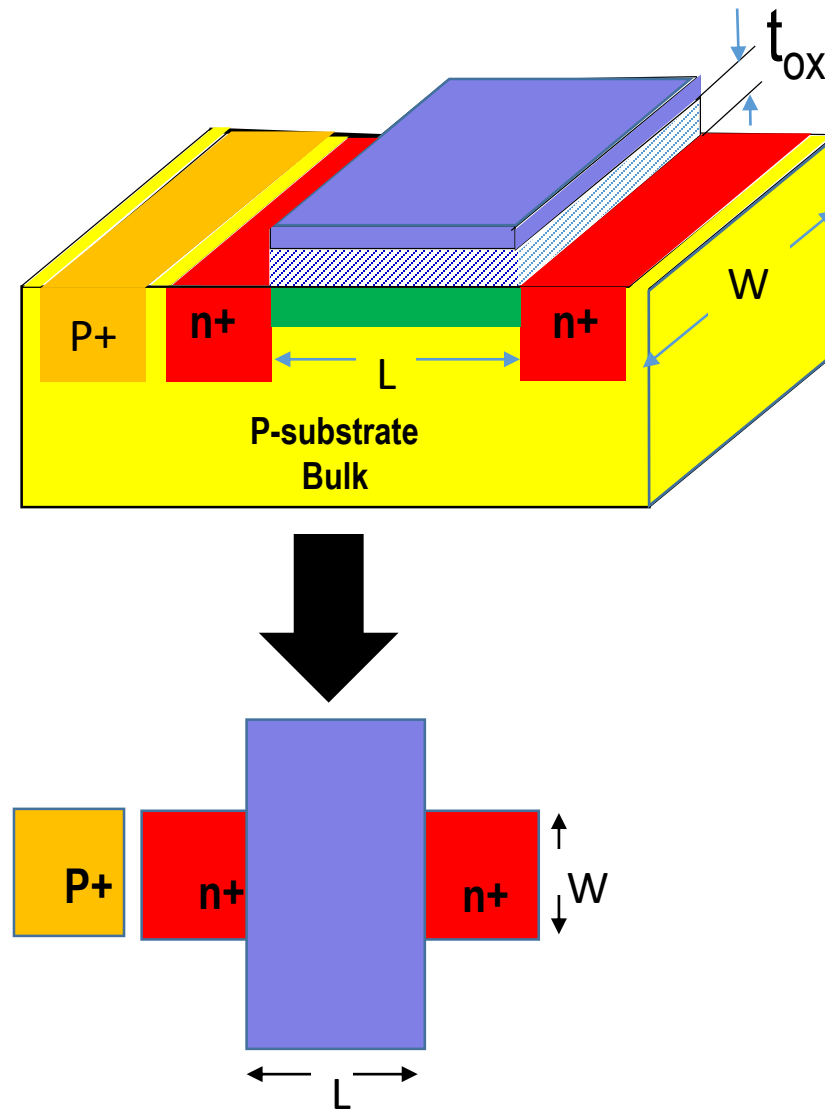
L : length of transistor

ϵ_0 : Permittivity of free space

ϵ_r : Relative permittivity of SiO₂

t_{ox} : Thickness of SiO₂

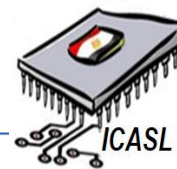
E- 3-D construction of transistor





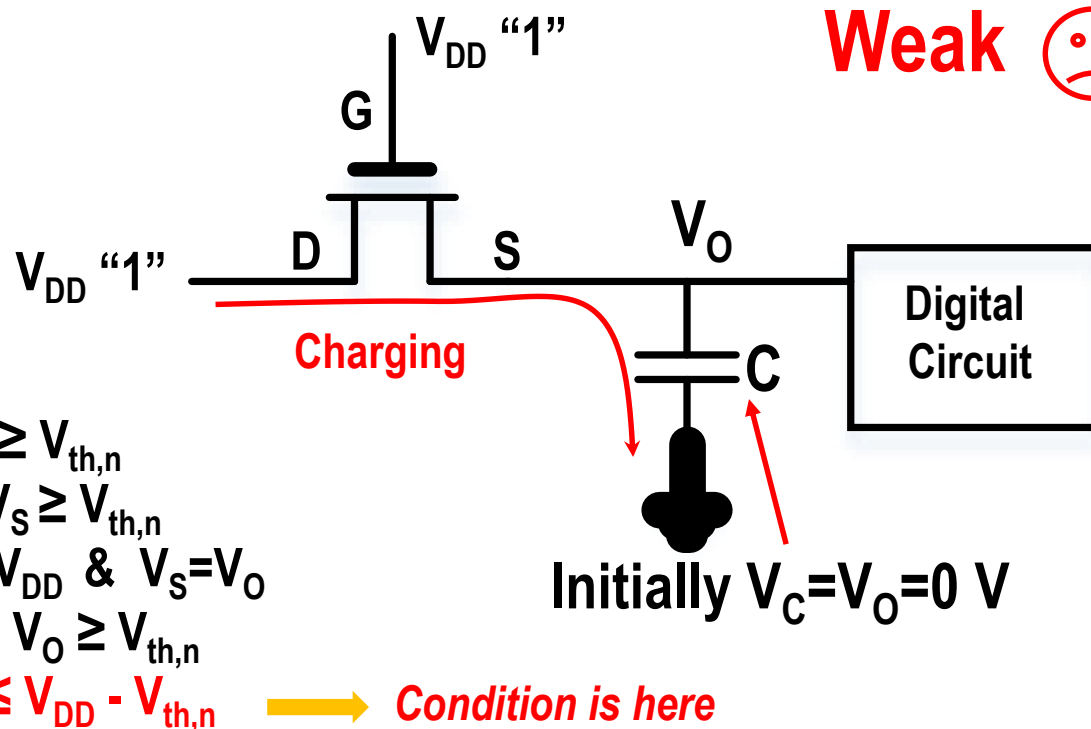
- ❑ Classification of transistors
- ❑ Enhancement MOSFET
- ❑ Ch/chs of E-MOSFET as a switch
- ❑ Synthesize of CMOS digital circuit

Ch/chs of E-MOSFET as a switch (1/5)

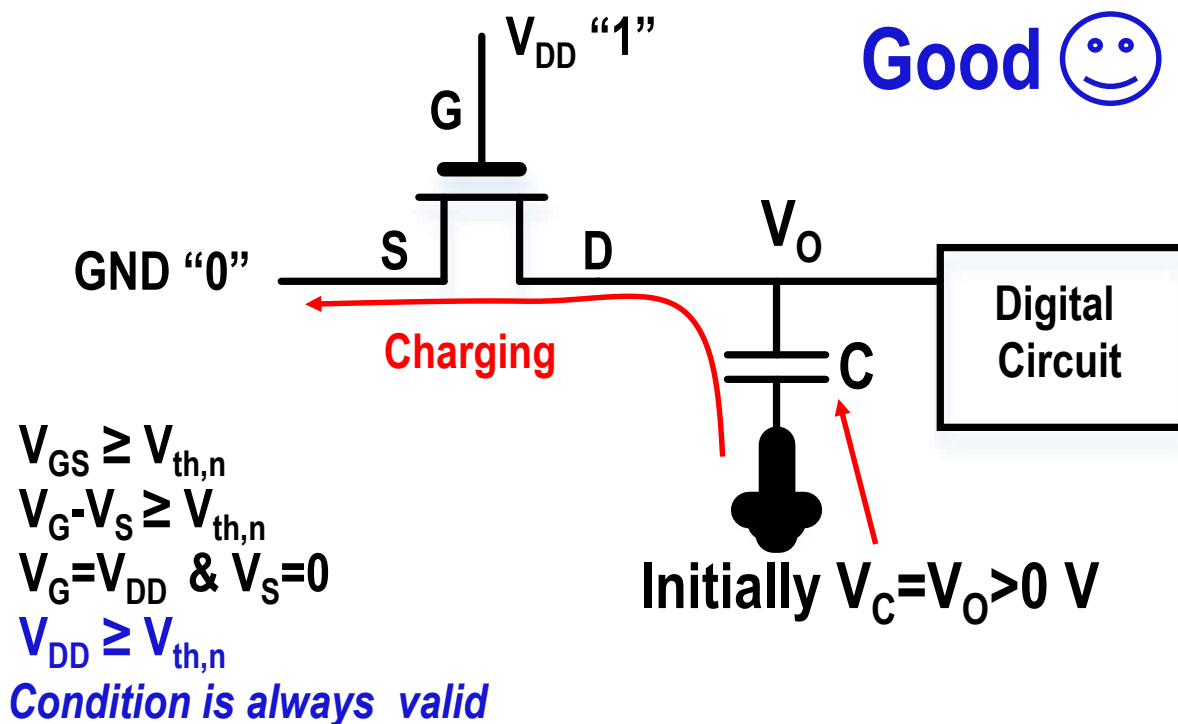


n-channel

Transfer "1"



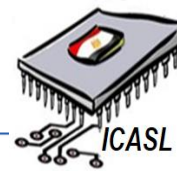
Transfer "0"



- ❑ When $V_G = V_{DD}$, n-channel E-MOSFET is **ON**.
- ❑ The capacitor starts charging and V_O increases till it reaches $V_{DD} - V_{th,n}$
- ❑ It means that Maximum value of V_O equals $(V_{DD} - V_{th,n}) \neq V_{DD}$

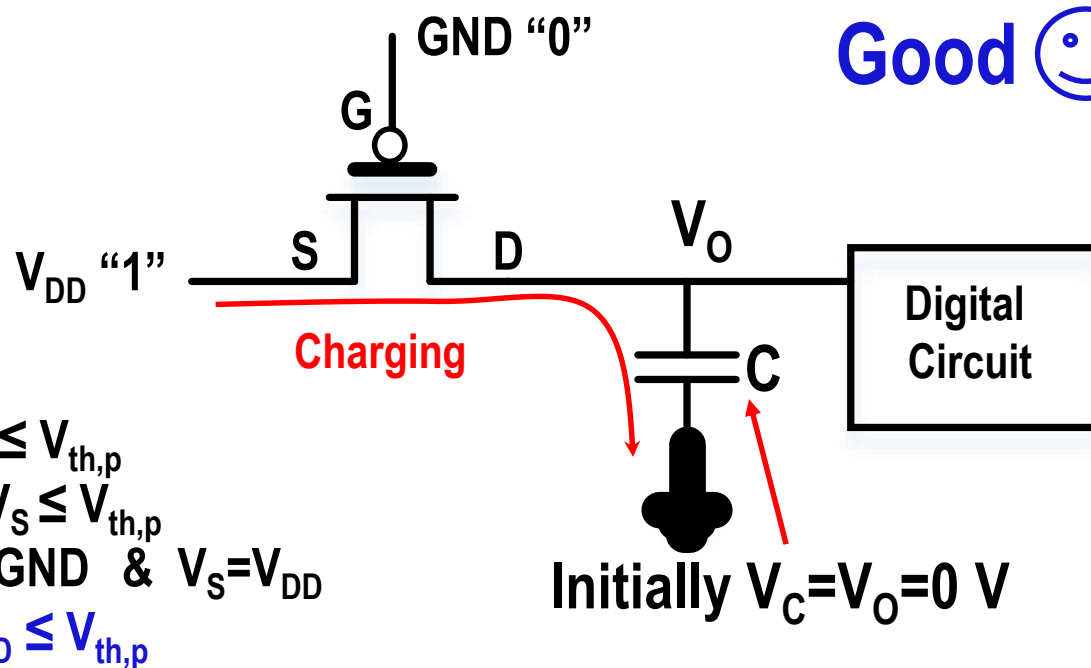
- ❑ When $V_G = V_{DD}$, n-channel E-MOSFET is **ON**.
- ❑ The capacitor starts discharging and V_O decreases till it reaches GND.
- ❑ It means that minimum value of V_O equals GND "0"

Ch/chs of E-MOSFET as a switch (2/5)



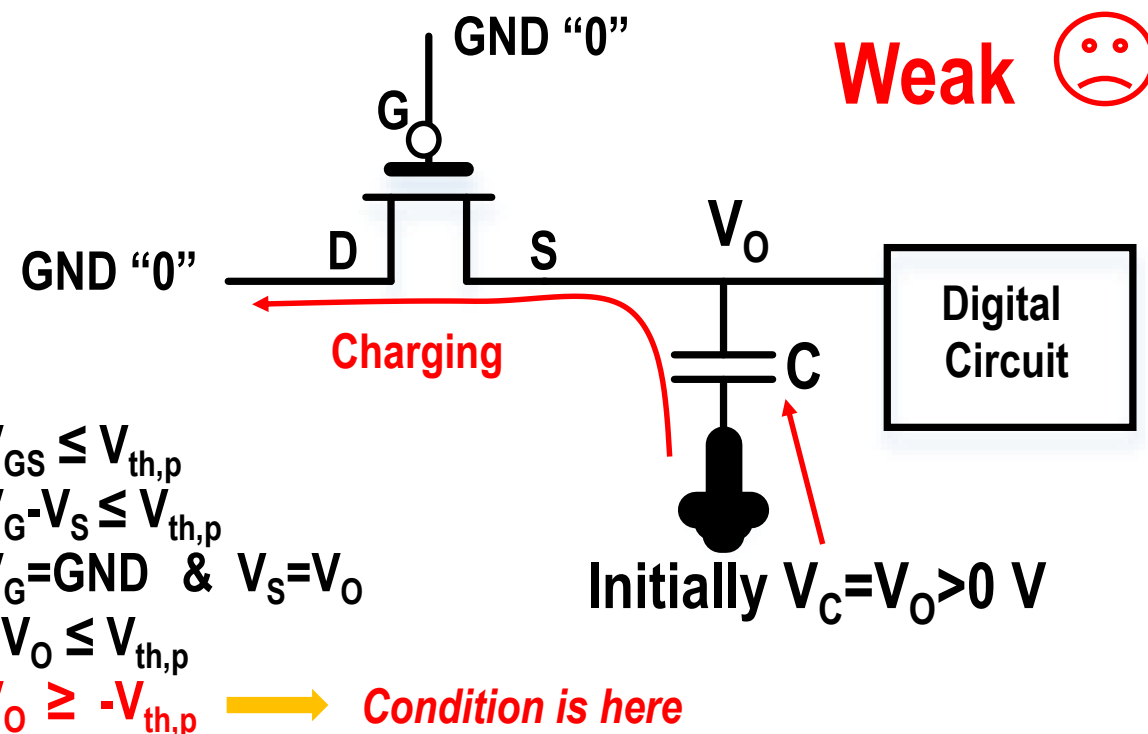
P-channel

Transfer "1"



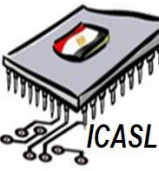
Condition is always valid

Transfer "0"



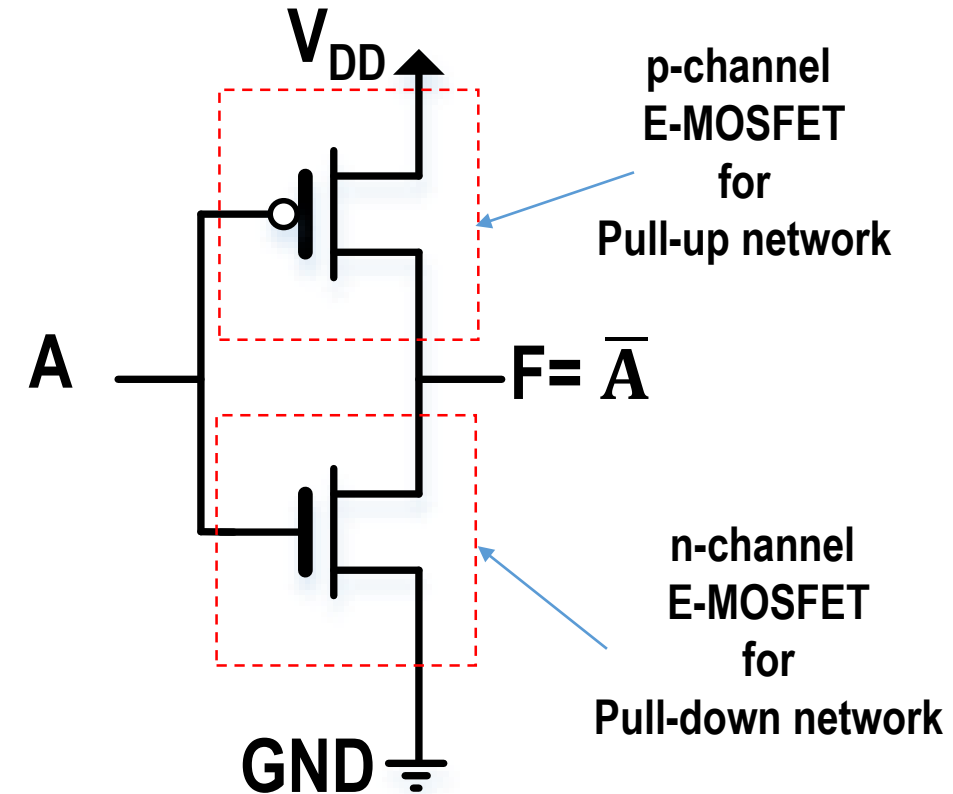
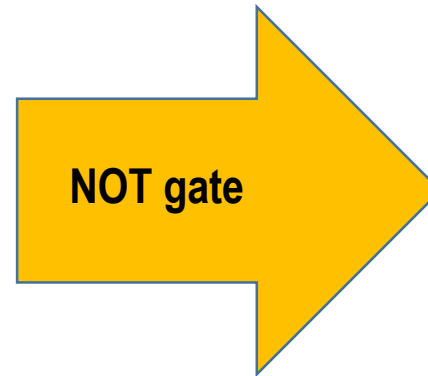
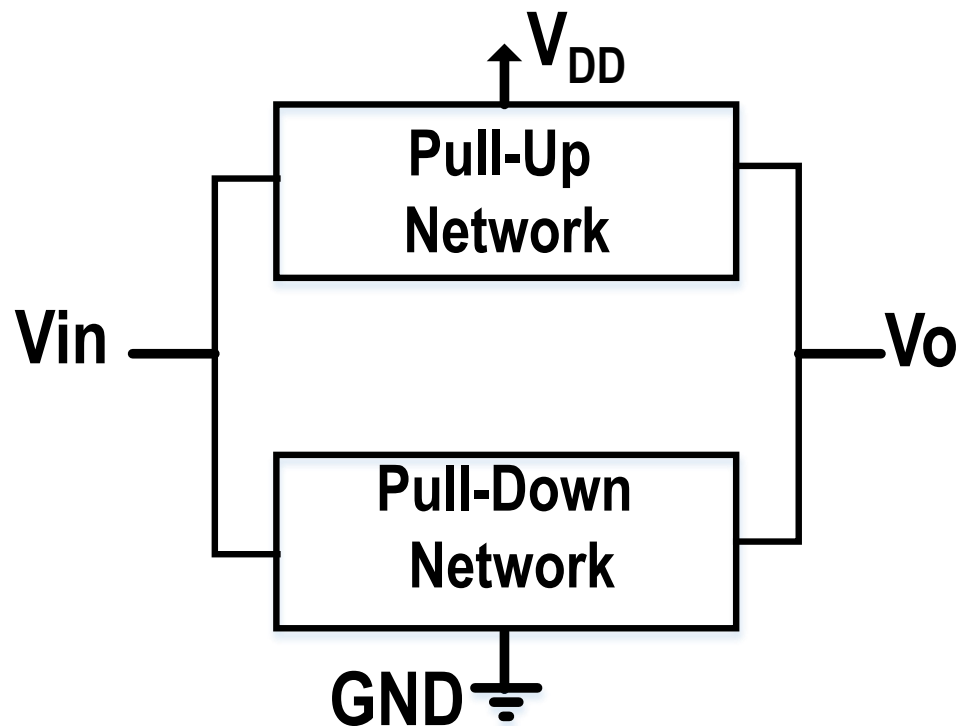
- ❑ When $V_G = \text{GND}$, p-channel E-MOSFET is **ON**.
- ❑ The capacitor starts charging and V_O increases till it reaches V_{DD}
- ❑ It means that maximum value of V_O equals V_{DD}
- ❑ When $V_G = \text{GND}$, p-channel E-MOSFET is **ON**.
- ❑ The capacitor starts discharging and V_O decreases till it reaches $-V_{th,p} = |V_{th,p}| \neq \text{GND}$
- ❑ It means that minimum value of V_O equals GND "0"

Ch/chs of E-MOSFET as a switch (3/5)

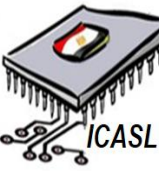


Note:-

	Good	Weak
n-channel EMOSFET	"0"	"1"
p-channel EMOSFET	"1"	"0"

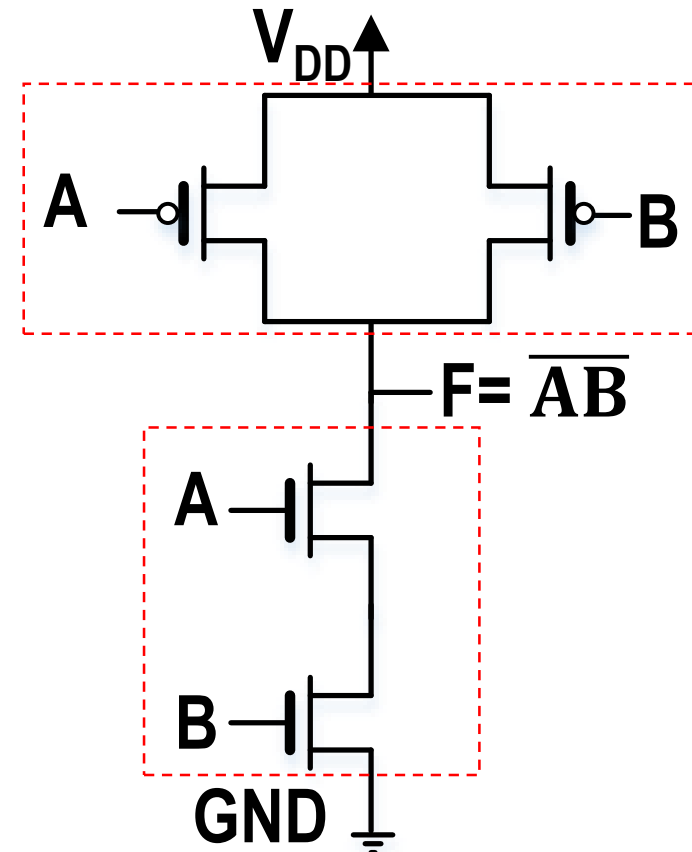
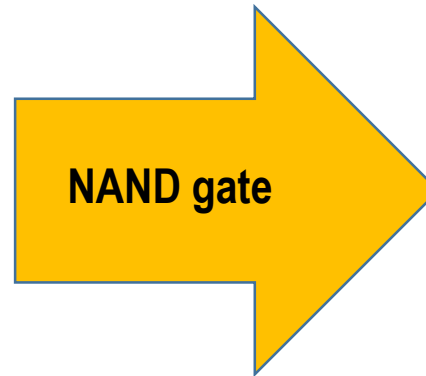
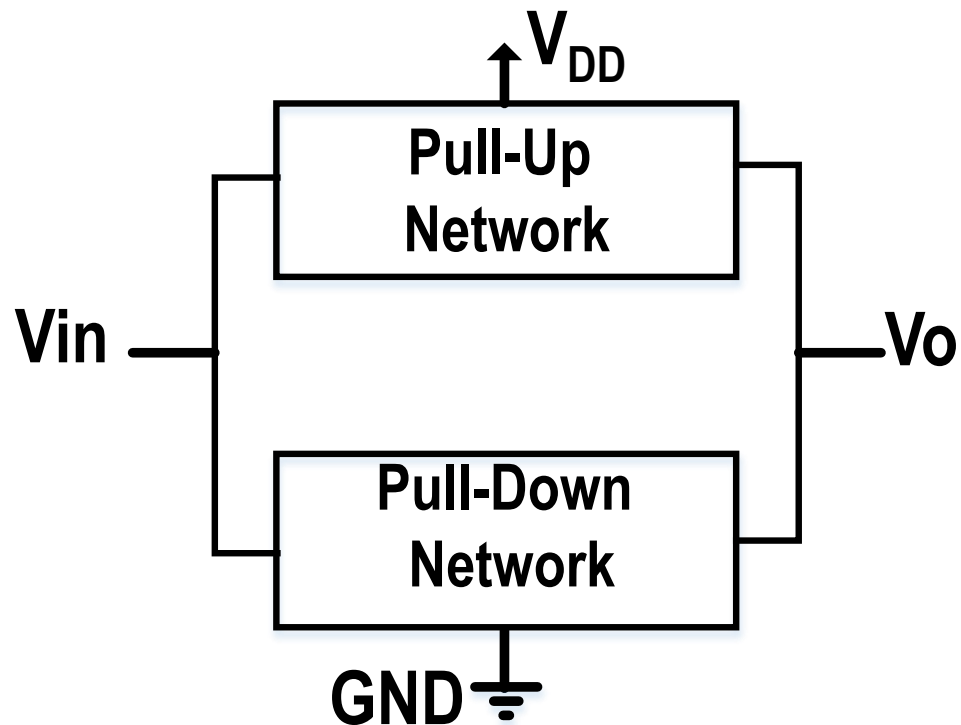


Ch/chs of E-MOSFET as a switch (4/5)



Note:-

	Good	Weak
n-channel EMOSFET	"0"	"1"
p-channel EMOSFET	"1"	"0"

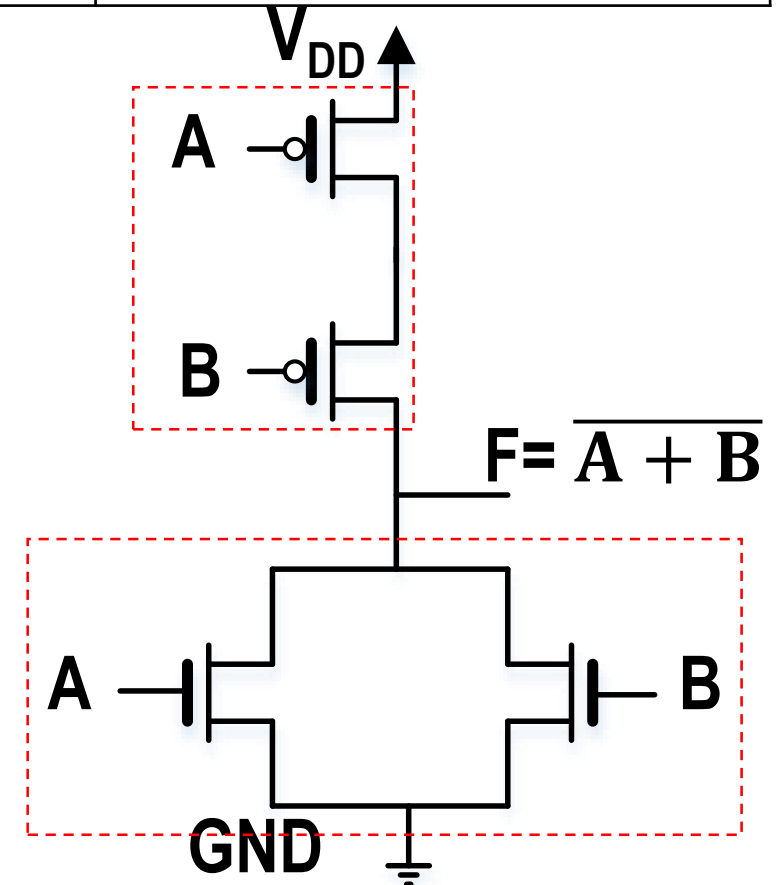
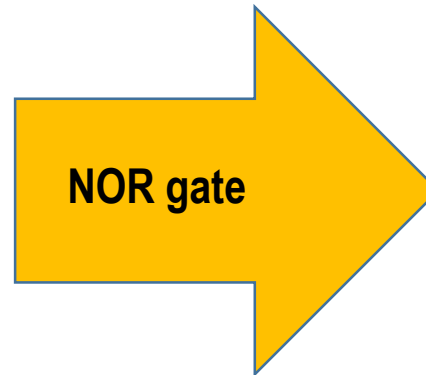
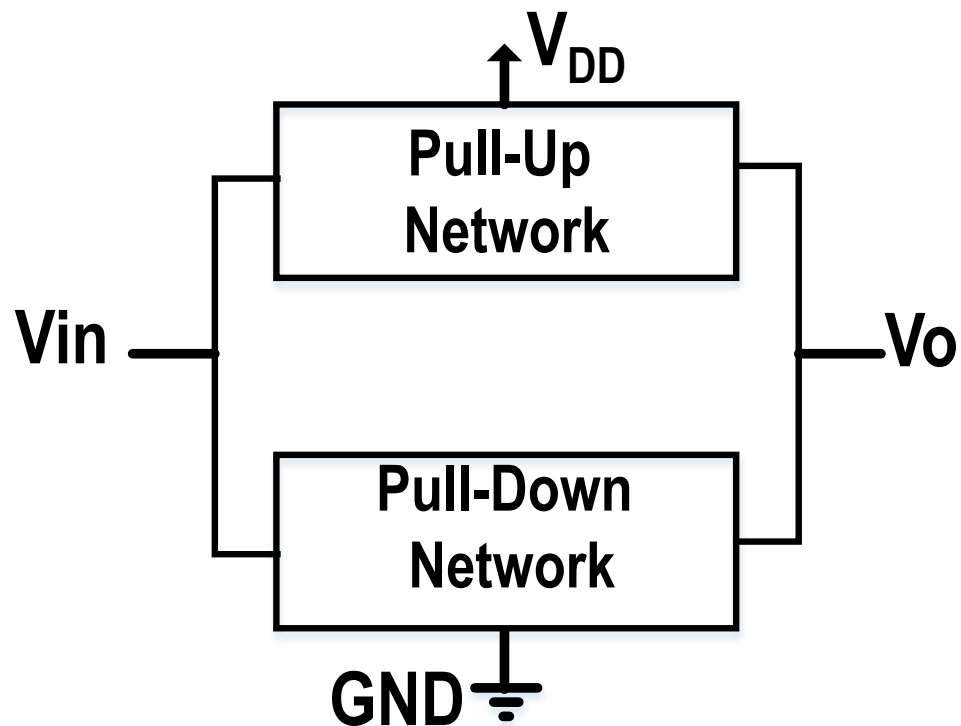


Ch/chs of E-MOSFET as a switch (5/5)



Note:-

	Good	Weak
n-channel EMOSFET	"0"	"1"
p-channel EMOSFET	"1"	"0"





- ❑ Classification of transistors
- ❑ Enhancement MOSFET
- ❑ Ch/chs of E-MOSFET as a switch
- ❑ Synthesize of CMOS digital circuit

Synthesize of CMOS digital circuit (1/11)

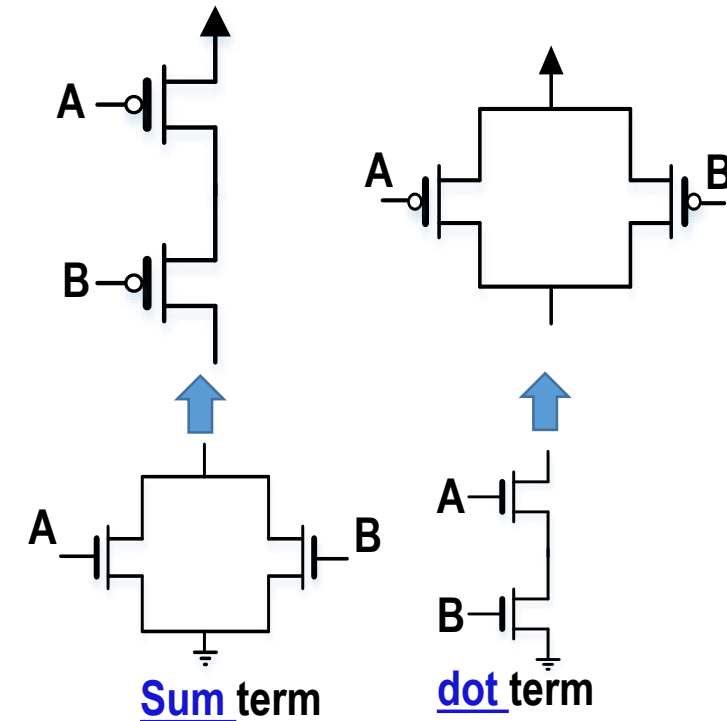


In order to synthesize or implement a CMOS digital circuit for logic functions using CMOS transistors. You have to follow these steps.

- 1- Simplify the function by Boolean algebra or K-map
- 2- Get $\bar{F}$ by collecting "0" as a form of sum of product (SOP)
- 3- Firstly, draw the pull-down network by using n-channel E-MOSFET.
 - 3.1 Any dot term will be drawn as series connection. Ex, $\bar{F} = A.B$
 - 3.2 Any sum term will be drawn as parallel connection. Ex. $\bar{F} = A+B$

↓
Vice versa
↓

- 4- Secondly, draw the pull-up network using p-channel E-MOSFET.
 - 4.1 **Parallel** connection is converted into series connection.
 - 4.2 **Series** connection is converted into parallel connection



Synthesize of CMOS digital circuit (2/11)



Ex.1. Synthesize $F = \overline{A \cdot B}$ using CMOS transistors and count the total number of transistors.

Ans.

Setp1: $F = \overline{A \cdot B}$ is simplified

Setp2: $\overline{F} = A \cdot B$

Setp3: **Firstly**, draw the **pull-down** network

□ Two transistors {A,B} in series connection

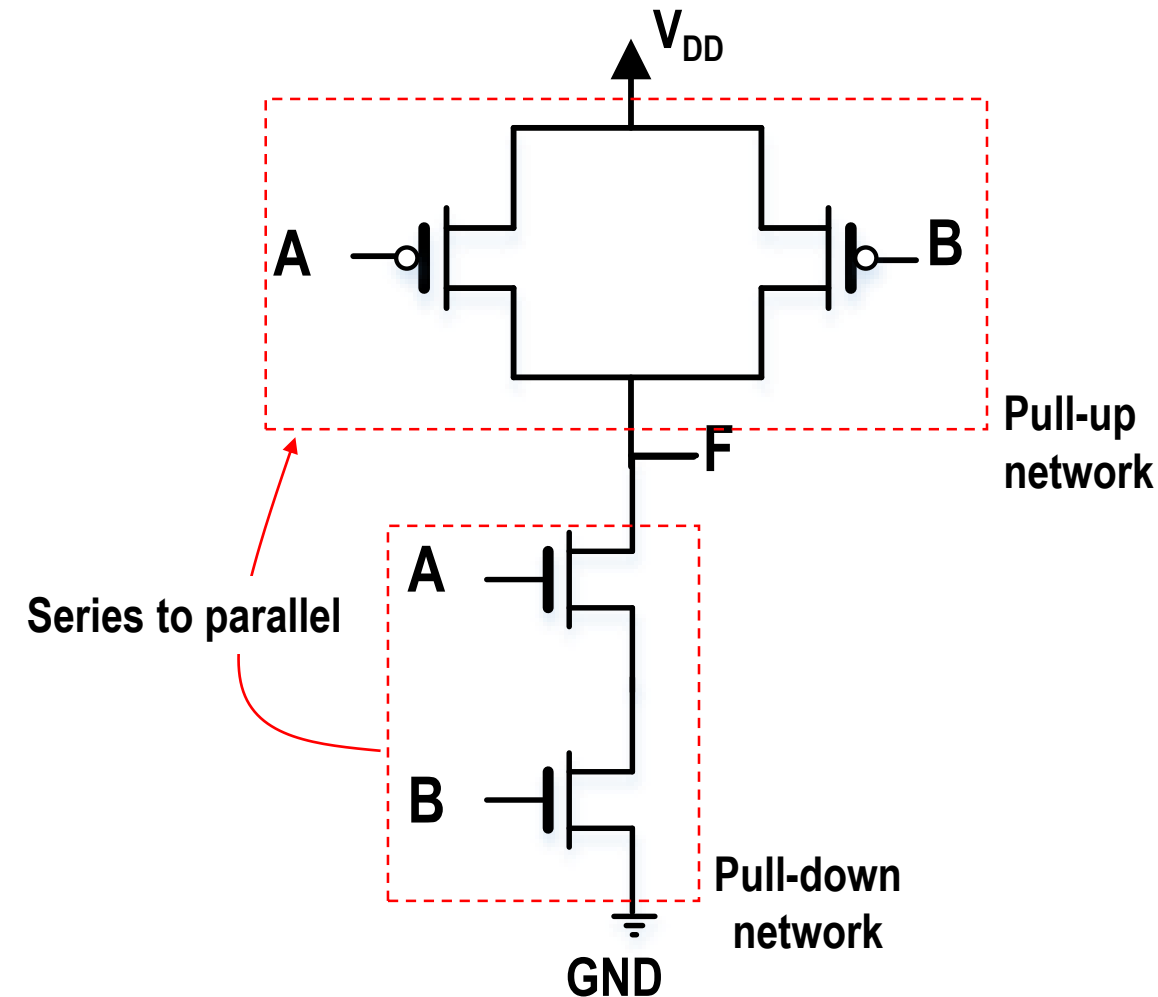
Setp4: **secondly**, draw the **pull-up** network

□ Two transistors {A,B} in parallel connection

of transistors = $2 \cdot 2 = 4$ 😊



of transistors in the pull-down transistor



Synthesize of CMOS digital circuit (3/11)



Ex.2. Synthesize $F = \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{D}$ using CMOS transistors and count the total number of transistors

Ans.

Setp1: $F = \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{D}$ is simplified by K-map

CD AB		C			
		00	01	11	10
A	00	1	1	0	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	0	0	1

Diagram illustrating the K-map for the function $F = \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{D}$. The K-map is a 4x4 grid with rows labeled AB (00, 01, 11, 10) and columns labeled C (00, 01, 11, 10). The values in the cells are 1, 1, 0, 1 for AB=00; 1, 0, 0, 1 for AB=01; 1, 0, 0, 1 for AB=11; and 1, 0, 0, 1 for AB=10. The map is simplified by grouping the 0s into three groups: a group of four 0s (AB=01, 11, 11, 10) labeled B, a group of four 0s (AB=01, 11, 11, 10) labeled D, and a group of four 0s (AB=01, 11, 11, 10) labeled C.

Setp2: $\overline{F} = A.D + B.D + C.D = D.(A+B+C)$

Synthesize of CMOS digital circuit (4/11)



Ex.2. Synthesize $F = \overline{A} \cdot \overline{B} \cdot \overline{C} + \overline{D}$ using CMOS transistors and count the total number of transistors

Ans.

Setp2: $\overline{F} = D \cdot (A+B+C)$

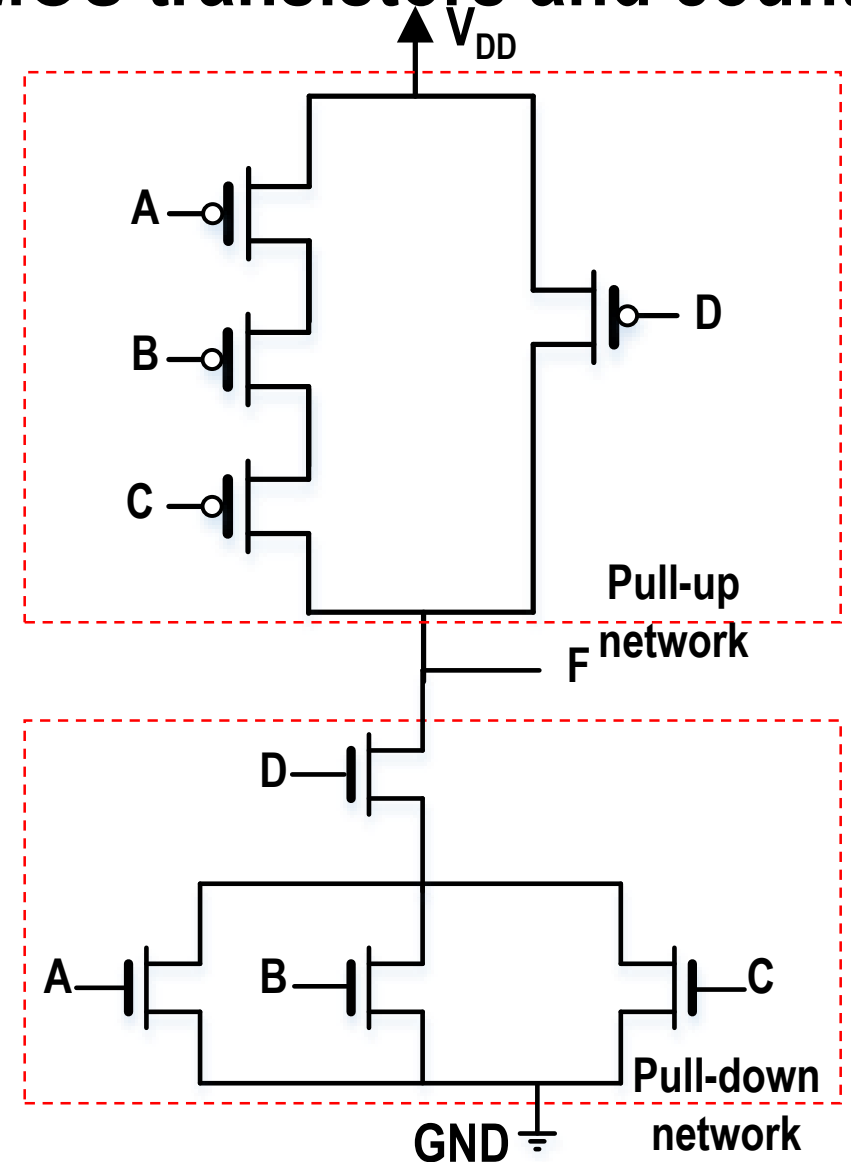
Setp3: **Firstly**, draw the **pull-down** network

- Three transistors {A,B,C} in parallel connection.
- All are connected in series with transistor {D}.

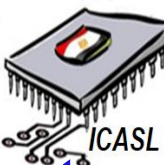
Setp4: **Secondly**, draw the **pull-up** network

- Three transistors {A,B,C} in series connection.
- All are connected in parallel with transistor {D}.

of transistors = $2 \times 4 = 8$ 😊



Synthesize of CMOS digital circuit (5/11)



In order to synthesize or implement a CMOS digital circuit for logic functions using **Universal gates** (**NAND, NOR and NOT**).

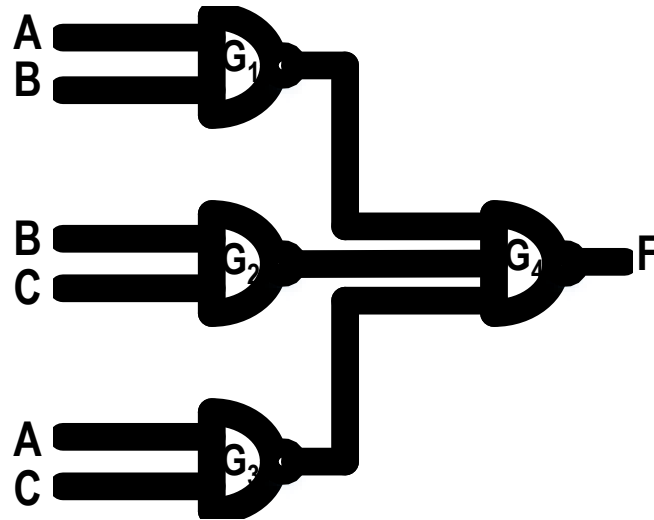
NAND Gate

❑ The function must be in form of Sum Of Product (SOP)

Ex: $F = AB + BC + AC$

$$\overline{\overline{F}} = \overline{AB + BC + AC}$$
$$\overline{\overline{F}} = \overline{AB} \cdot \overline{BC} \cdot \overline{AC}$$

$\underbrace{\quad}_{G_1} \quad \underbrace{\quad}_{G_2} \quad \underbrace{\quad}_{G_3}$
 $\underbrace{\hspace{10em}}_{G_4}$



of transistors = $3 * 2 * 2 + 1 * 3 * 2 = 18$ 😊

NAND gate

of terminal/gate

for pull-up and down networks

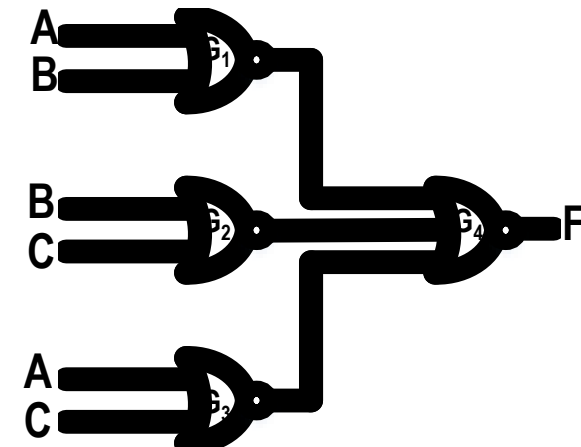
NOR Gate

❑ The function must be in form of Product of Sum (POS)

Ex: $F = (A+B) \cdot (B+C) \cdot (A+C)$

$$\overline{\overline{F}} = \overline{(A+B) \cdot (B+C) \cdot (A+C)}$$
$$\overline{\overline{F}} = \overline{(A+B)} + \overline{(B+C)} + \overline{(A+C)}$$

$\underbrace{\quad}_{G_1} \quad \underbrace{\quad}_{G_2} \quad \underbrace{\quad}_{G_3}$
 $\underbrace{\hspace{10em}}_{G_4}$



of transistors = $3 * 2 * 2 + 1 * 3 * 2 = 18$ 😊

Synthesize of CMOS digital circuit (6/11)



Ex. Synthesize $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ using
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) CMOS transistors

Setp1: $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ is simplified by K-map

		B			
		00	01	11	10
A	0	1	0	1	1
	1	1	1	0	1

BC is written above the first column. A is written to the left of the first column. Brackets indicate groupings: a horizontal bracket over the top row (B=01, 11) and a vertical bracket to the right of the bottom row (A=1).

Setp2: $\overline{F} = A \cdot B \cdot C + \overline{A} \cdot \overline{B} \cdot C = C \cdot (A \cdot B + \overline{A} \cdot \overline{B})$

Synthesize of CMOS digital circuit (7/11)



Ex. Synthesize $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ using

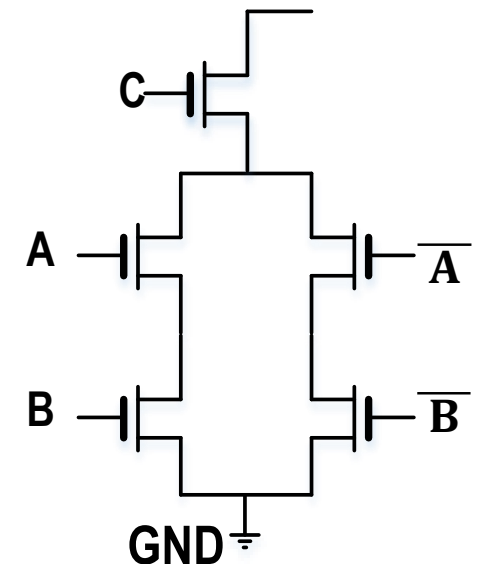
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) CMOS transistors

Setp2: $\overline{F} = C \cdot (A \cdot B + \overline{A} \cdot \overline{B})$

Setp3: **Firstly**, draw the **pull-down** network

- ☐ Two transistors {A,B} in series connection
- ☐ Two transistors $\{\overline{A} \cdot \overline{B}\}$ in series connection.
- ☐ {A,B} and $\{\overline{A} \cdot \overline{B}\}$ are connected in parallel connection.
- ☐ Transistor {C} is in series connection with {A,B} and $\{\overline{A} \cdot \overline{B}\}$



Synthesize of CMOS digital circuit (8/11)



Ex. Synthesize $F = A \cdot \overline{B} + \overline{A} \cdot B + \overline{C}$ using

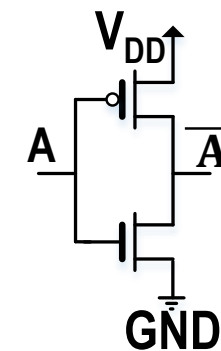
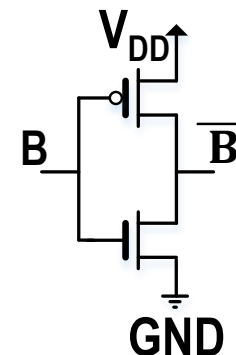
(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.
(a) CMOS transistors

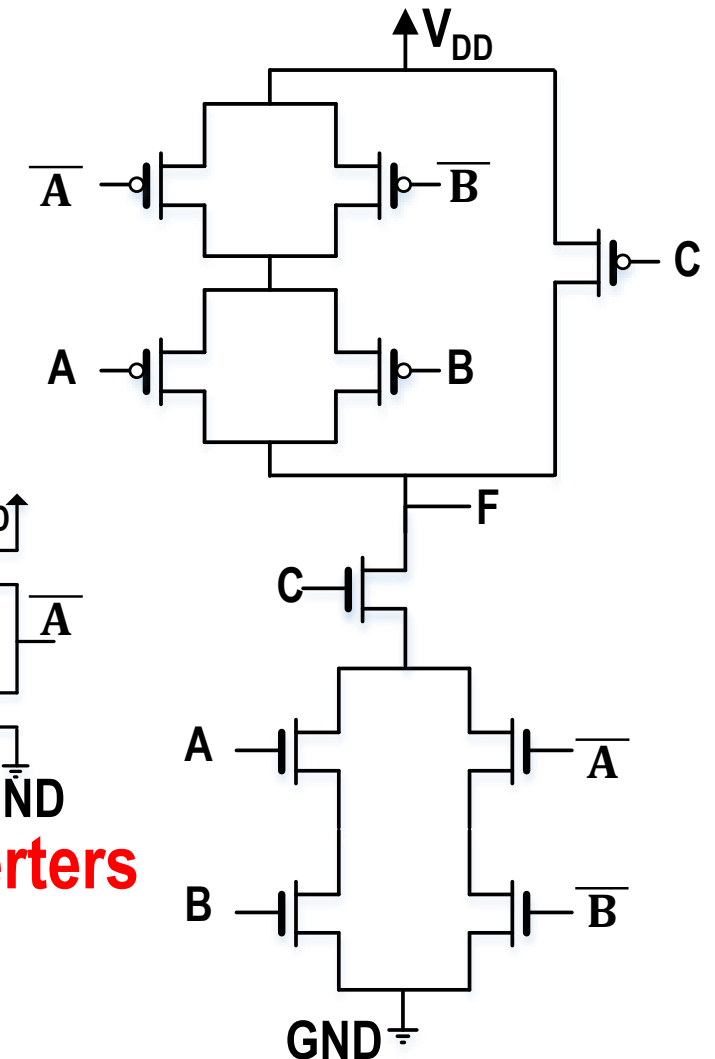
Setp2: $\overline{F} = C \cdot (A \cdot B + \overline{A} \cdot \overline{B})$

Setp4: **Secondly**, draw the **pull-up** network

- ❑ Two transistors {A,B} in parallel connection
- ❑ Two transistors $\{\overline{A} \cdot \overline{B}\}$ in parallel connection.
- ❑ {A,B} and $\{\overline{A} \cdot \overline{B}\}$ are connected in series connection.
- ❑ Transistor {C} is in parallel connection with {A,B} and $\{\overline{A} \cdot \overline{B}\}$



Don't Forget the inverters



of transistors = $2 \cdot 5 + 2 \cdot 2 = 14$ 😊

Synthesize of CMOS digital circuit (9/11)



Ex. Synthesize $F = A \cdot \bar{B} + \bar{A} \cdot B + \bar{C}$ using

(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

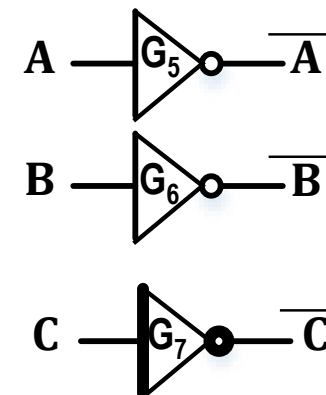
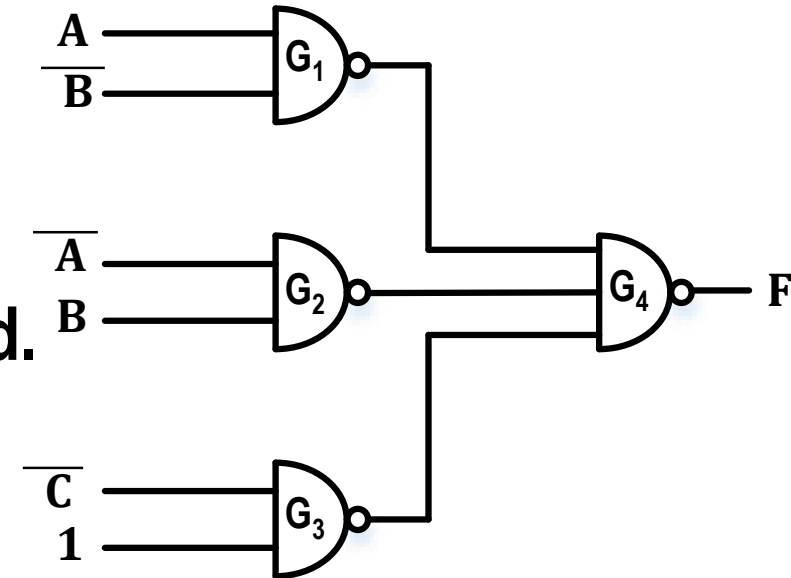
Ans.
(a) Universal gates

Due to the function is in form of **SOP**, NAND gates are used.

$$\overline{\overline{F}} = \overline{A \cdot \bar{B} + \bar{A} \cdot B + \bar{C} \cdot 1}$$

$$\overline{\overline{F}} = \underbrace{\overline{A \cdot \bar{B}}}_{G_1} \cdot \underbrace{\overline{\bar{A} \cdot B}}_{G_2} \cdot \underbrace{\overline{\bar{C} \cdot 1}}_{G_3}$$

$$\quad \quad \quad \underbrace{\quad \quad \quad}_{G_4}$$



of transistors = $3 \cdot 2 \cdot 2 + 1 \cdot 3 \cdot 2 + 3 \cdot 2 = 24$ 😊

Synthesize of CMOS digital circuit (10/11)



Ex. Synthesize $F = A \cdot \bar{B} + \bar{A} \cdot B + \bar{C}$ using

(a) CMOS transistors (b) Universal gates (c) Compare the used transistors in (a) and (b)

Ans.

(C) Compare the used transistors in (a) and (b)

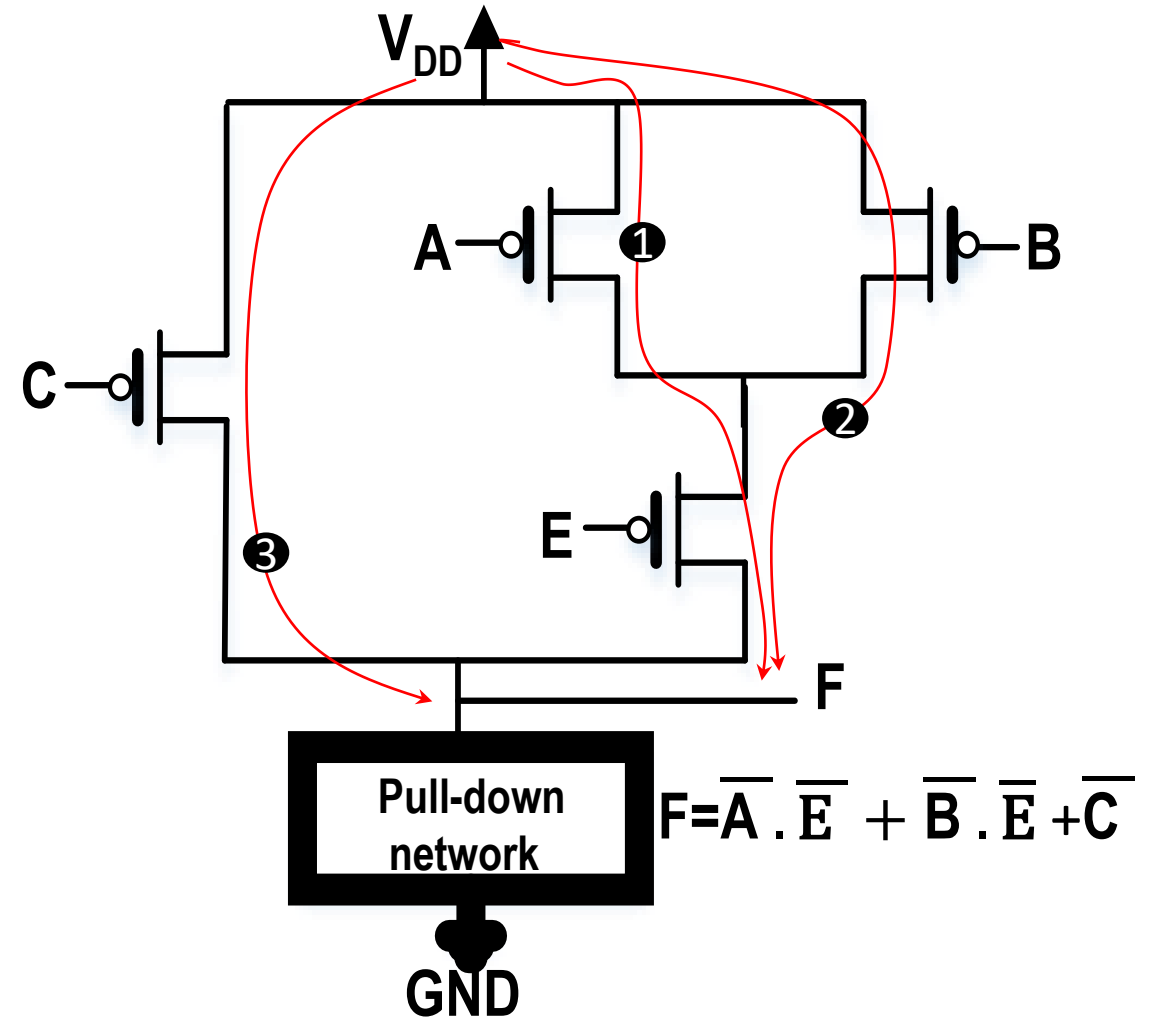
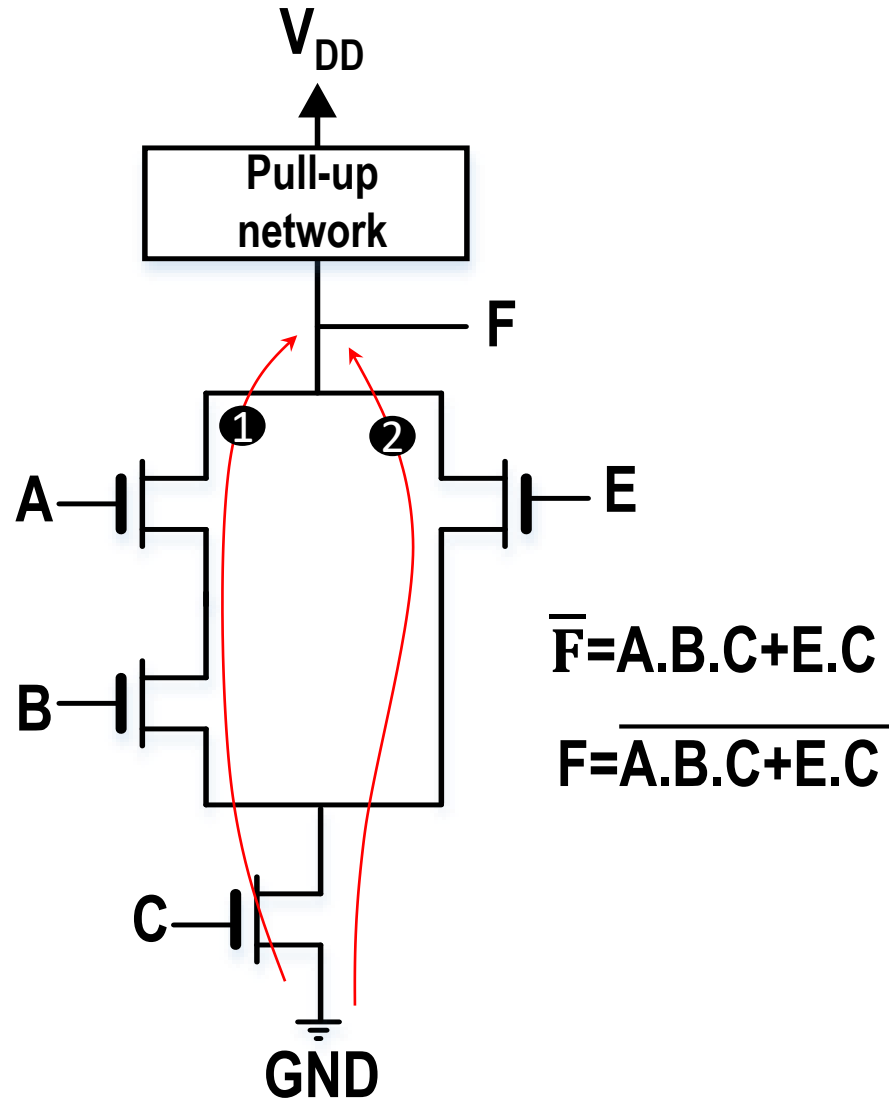
	CMOS transistors	Universal gates
# of transistors	14 😊	24 ☹️

Synthesizing by **CMOS transistors** requires low number of transistors,
Thus, the consumption power is low and the consumed silicon area is small

Synthesize of CMOS digital circuit (11/11)



Find the transfer function of the following circuit



Thanks!